REVISION 2021 • PRINTING TECHNOLOGY • SEMESTER 4

Digital Imaging Techniques

A professional diploma handbook for Course Code 4103 covering digital file formats, PDF and JDF, boxes and marks, RIP, data transfer, workflow software, CTP, digital printing, inkjet, laser, LED and direct imaging technology.

4103
Course code

4
Credits

60
Periods

Program Core
Category

Digital imaging □ file format, PDF/X, RIP, CTP, workflow, inkjet, laser □□□□□□□ print quality, colour accuracy, output speed, waste control □□□□□□□□□ □□□□□□□□□ □□□□□□□□□□□□□□. Exam answer-□ workflow diagram □□□□□□□□□□□□ □□□□□□□□□□.

Digital prepress to print flow

Course overview

Objectives

Provide cutting-edge knowledge in digital imaging, keep pace with international changes in printing and graphic arts, and expose students to digital prepress and digital printing applications.

Prerequisites

Bit, byte and bitmap images; WYSIWYG and colour theories; types of images and originals from earlier printing courses.

Course outcomes

CO	Outcome	Hours
CO1	Explain file formats, proofing, PDF production and management.	15
CO2	Interpret data transfer by using computer.	14
CO3	Summarize modern digital printing technologies.	15
CO4	Outline inkjet and laser printing technology.	14

Source / syllabus information

Item	Official information
Programme	Diploma in Printing Technology
Code / title	4103 - Digital Imaging Techniques
Semester / credits	Semester 4 / Credits 4
CO-PO mapping	CO1 to CO4 moderately mapped to PO1.
Model paper	Not specified in supplied syllabus/source.

Redesigned syllabus roadmap

M1 File, PDF, RIP

Draw PDF box diagram, RIP stages and proofing flow. Explain raster/vector file formats, PDF, JDF and marks.

M2 Workflow and CTP

Explain automated workflow, production management software, preflight, job ticket and UV/Thermal CTP.

M3 Digital printing

Explain digital printing workflow, advantages, pixels, bit depth, electro photography, ionography, thermography, toner jet and inks.

M4 Inkjet / laser

Explain DOD, continuous inkjet, binary/multi deflection, thermal, piezo, laser, LED, substrates and direct imaging.

Module 1 - File Formats, PDF, JDF, RIP and Proofing

CO1 • 15 hours • Digital prepress foundation.

File formats

JPEG/JPG is compressed raster, TIFF is high-quality raster, PNG supports transparency, GIF supports limited colour/animation, BMP is uncompressed bitmap, PSD is Photoshop, CDR is CorelDraw, INDD is InDesign, PDF is exchange/output format and XML/JDF carry structured workflow data.

Raster vs vector

Raster files are pixel-based and resolution dependent. Vector files are mathematical path-based and scale without pixelation. Printing jobs often combine both.

PDF and PostScript

PDF is device-independent document exchange format with fonts, images and layout. PostScript is page description language. PDF is easier for viewing/transfer; PostScript is interpreted by output devices.

Boxes and marks

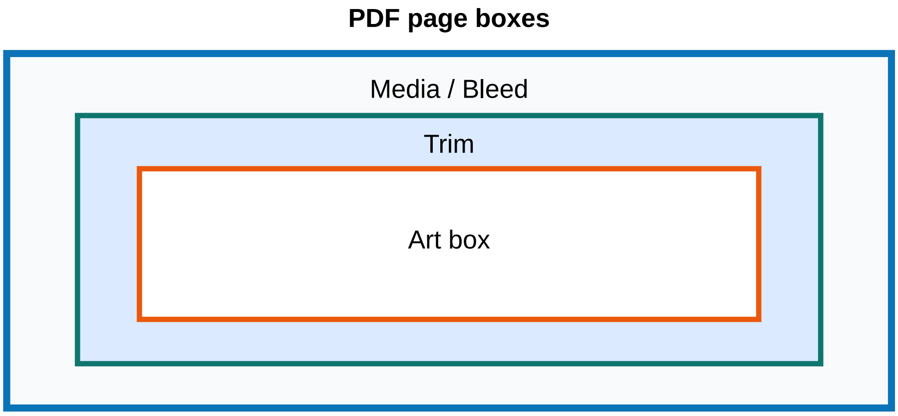
Media box, trim box, bleed box and art box define page geometry. Trimming marks, cutting marks, registration marks and quality control patches guide accurate production.

RIP

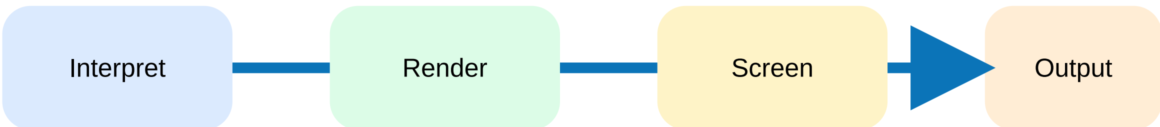
Raster Image Processor converts page description into screened raster data for output. Stages include interpretation, rendering, screening and output.

Proofing

Digital proofing verifies content, colour and layout before final output. Digital halftone proof simulates final screened result.



PDF boxes for digital document setting.



RIP stages.

Differentiate raster and vector formats.

Explain PDF boxes and print marks.

Explain RIP and its stages.

Module 2 - Workflow Software, Job Tickets and CTP

CO2 • 14 hours • Data transfer and computer-based production.

Modern print workflow

Automated workflow connects job entry, preflight, imposition, proofing, RIP, plate/output, printing and finishing. It reduces manual errors and tracks status.

Workflow software

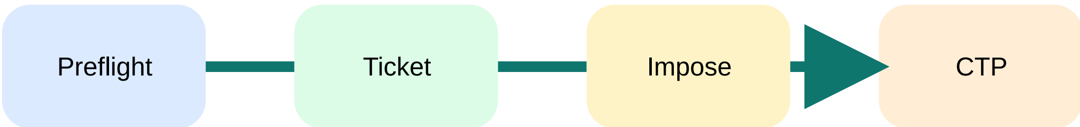
Production management software and systems such as Prinect use modules for signastation, meta dimension and production control. Preflight checks missing fonts, low-resolution images, wrong colour spaces and bleed errors.

Job tickets

A job ticket stores customer, job size, quantity, substrate, colour, imposition, finishing, proofing and delivery information. Digital job ticket based workflow automates routing.

CTP

Computer-to-plate transfers digital data directly to plate. UV and thermal CTP systems expose compatible plates and improve registration, speed and consistency.



Module 3 - Digital Printing Technologies

CO3 • 15 hours • Technology, workflow, applications and inks.

Digital printing

Digital printing prints directly from digital file without conventional plate for each job. It supports variable data, short runs, fast turnaround and on-demand production.

Digital image

A pixel is the smallest picture element. Bit depth defines number of tonal or colour levels. Higher bit depth improves gradation but increases file size.

Technologies

Electrophotography, ionography, thermography, toner jet, magnetography, dot matrix and inkjet are important digital/non-impact or impact technologies.

Inks

Toner inks are dry or liquid particles fused to substrate. Inkjet inks may be dye, pigment, solvent, UV, latex or solid ink depending on substrate and application.

Module 4 - Inkjet, Laser, LED, Substrates and Direct Imaging

CO4 • 14 hours • Output engines and comparison.

Inkjet printing

Inkjet forms image by ejecting droplets. Drop-on-demand ejects droplets only where needed; continuous inkjet generates continuous stream and deflects drops.

Inkjet principles

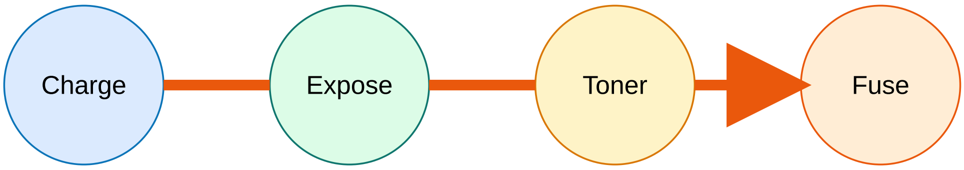
Binary deflection, multi-deflection, thermal inkjet, piezo inkjet, electrostatic inkjet and solid ink are important methods. Cartridge and print head condition affects quality.

Laser and LED

Laser printer uses electrophotographic process: charge, expose, develop, transfer, fuse and clean. LED printer uses LED array instead of laser scanning system.

Substrates and direct imaging

Digital printing substrates include paper, board, plastics, cloth, metal foils/sheets, glass and ceramic. Direct imaging presses image plates on press and reduce setup time.



Rule bank / definition bank

PDF/X rule

Use print-ready PDF standards for reliable exchange, embedded fonts, correct colour and predictable output.

RIP rule

If RIP settings are wrong, output can fail even when layout appears correct on screen.

Inkjet rule

Droplet size, print head condition, substrate absorption and ink chemistry control image quality.

Practice and quick revision

One-mark: Define RIP.

Short answer: Differentiate raster and vector formats.

Three-mark: Explain PDF boxes.

Descriptive: Explain automated digital print workflow.

Application: Select digital printing method for variable data labels.

Objective questions

1. RIP expands to

A. Raster Image Processor B. Rapid Ink Plate C. Register Image Point D. Raster Ink Proof

Answer: A

2. A vector file is mainly based on

A. pixels B. paths C. solvent D. toner

Answer: B

Fresh model question paper

Course Code: 4103
Course: Digital Imaging Techniques
Time: 3 Hours | Maximum Marks: 100

Part A
1. Define raster image.
2. Name four file formats used in printing.
3. What is PDF?
4. Define JDF.
5. What is RIP?
6. Define preflight.
7. What is CTP?
8. Define digital printing.
9. What is bit depth?
10. Define inkjet printing.

Part B

11. Differentiate raster and vector file formats.
12. Explain PDF boxes and print marks.
13. Explain PDF, PostScript and JDF in print workflow.
14. Explain RIP stages with diagram.
15. Explain modern print workflow and workflow software.
16. Explain job tickets and digital job-ticket based workflow.
17. Explain CTP systems and plates.
18. Explain digital printing technologies and applications.
19. Compare impact and non-impact printing.
20. Explain inkjet and laser printer working.

Part C

21. Prepare a digital workflow for a short-run colour brochure.
22. Compare digital printing and offset printing for variable-data jobs.

Complete answer guide

Answer format

Use definition, workflow diagram, file examples, device/application, advantages, limitations and common fault. For process answers, draw block diagram.

Application answer

For short-run colour brochure: collect artwork, preflight, convert to print-ready PDF/PDF-X, impose, proof, RIP, output by digital printer or CTP+press, finish and inspect.

References

Heidelberg Expert Guide PDF Workflow; Michael Bernard and John Peacock, Handbook of Printing and Production; Helmut Kipphan, Handbook of Print Media; Reid Anderson, Exploring Digital Prepress; Adams J.M., Printing Technology; Harald Johnson, Mastering Digital Printing.